

# Software Ground School

## WEEK 1:

What kind of software and hardware will be on our drone?

- NVIDIA Jetson Orin Nano is a computer meant to be deployed on robots
- Wifi antenna, GPS receiver x 2, cube(flight controller), 360 auto stabilize gimbal camera, other external sensors

Full software and Hardware spec list

- Object detection
  - OpenCV (traditional computer vision library)
  - YOLO (manually supervised learning)
- Aerial Imagery/Stitching (mapping and surveying)
  - A whole bunch of theory surrounding it, people have been perfecting this for a while
  - We tried OpenCV but had better results with OpenDroneMap so farFlight Controller
  - Hardware
    - Cube orange+, Pixhawk
  - Firmware
    - ArduPilot, Betaflight, PX4
  - Keeps the drone stable
  - Has very little autonomy (doesn't know what to do unless it takes command)
  - Takes commands via MAVLink over a physical link (UART serial cable) from the onboard computer
- Onboard computer, sensors, and middleware
  - jetson/pi running Ubuntu; main brain of the system
  - Runs code under the ROS framework that tells the flight controller what to do
  - The code takes flight controller and/or filtered sensor data (position, velocity, body rates) as input, processes these data using logic/algos you will implement, and generates fly commands. ROS will help you get data from and send commands to the hardware

What can I expect to be doing in the software team?

- Prioritize competitions
  - Time-based waypoint flying
  - mapping/surveying
  - Object detection, classification, and localization
  - Package drop/payload delivery

What is the state of drones in industry?

| feature        | Basic stack               | Production stack       |
|----------------|---------------------------|------------------------|
| Flight control | arduPilot,/PX4/BetaFLight | PX4 or ArduPilot on FC |

|                       |                                                       |                                                              |
|-----------------------|-------------------------------------------------------|--------------------------------------------------------------|
| Higher-level software | Python scripts, simple onboard computer apps, or none | ROS 2 or custom middleware on companion computer             |
| Autonomy              | Waypoint following                                    | Perception + object localization                             |
| Robustness & safety   | Basic failsafes                                       | Redundant sensing, health monitoring, extensive logging, etc |

BVLOS: Beyond Visual Line of Sight

How do we learn about drones effectively?

- Use AI
- Great for conceptual questions
- 1. Talk back and forth with AI to ensure conceptual understanding
- 2. Set goals for yourself, do projects(skill boosters)
- 3. Find ways of tracking progress
- 4. For individual details